

University of Engineering and Technology, Taxila
Department of Computer Science



Semester Project Proposal
Object Oriented Programming (CS-102-L)

Project Title:

Car Rental and Reservation System

Submitted by:

Wajeesh ul Hassan	25-CS-02
Hafiz M. Saeed	25-CS-05
M. Irfan Yasir	25-CS-194
Hamza Khurram	25-CS-128

Submitted To:

Sir Faheem Saleem

Lab Lecturer, Dept. of Computer Science

Semester: Spring 2026 | **Section:** B | **Date:**

Project Proposal

1. Executive Summary

This proposal outlines the development of an Automated Car Rental & Reservation System (ACRRS) developed in C++ using Object-Oriented Programming (OOP) paradigms. Unlike basic management systems, this project utilizes classes, inheritance, and encapsulation to create a robust, scalable environment for managing a car fleet. The system bridges the gap between manual vehicle tracking and modern digital logistics, providing a seamless interface for both administrators and customers. Key focus areas include data persistence through file handling and memory management using dynamic objects.

2. Problem Statement

Current manual car rental processes suffer from data redundancy, difficulty in tracking vehicle maintenance, and human error in billing calculations. Specific issues include:

Availability Conflicts: Tracking which cars are currently on the road vs. in the garage.

Insecure Records: Manual logs are prone to damage and unauthorized access.

Billing Inaccuracy: Calculating late fees and daily rates manually leads to financial discrepancies.

The ACRRS aims to solve these by centralizing data into a secure, object-managed environment.

3. Objectives

The main objectives of the project are:

- To implement a secure User Authentication System (Admin vs. Customer).
- To manage a dynamic fleet of vehicles with real-time status updates.
- To automate the billing process based on rental duration and car category (Economy, Luxury, SUV).
- To ensure Data Persistence so that car and user records remain saved after the program closes.

4. System Overview

Add a Car: Enter new vehicle details including Car ID, model, category (Economy/Luxury), and rental rate per hour.

Display Available Cars: Show a real-time list of all vehicles that are currently in the garage and ready for rent.

Search for a Car: Find a specific vehicle quickly by searching for its ID or model name.

Issue a Car: Assign a vehicle to a customer and automatically record the Start Time using the system clock.

Return & Bill Calculation: Record the return of a vehicle, capture the End Time, and automatically calculate the total bill based on the exact time used.

Manage Records: Save all car data and rental history into files so the information is never lost.

5. Implementation of OOP Paradigms

- **Inheritance:** A hierarchy starting from a Vehicle base class to derived Economy, Luxury, and SUV classes.
- **Encapsulation:** Protecting sensitive financial logic and user data within private class members.
- **Polymorphism:** Utilizing virtual functions to allow different vehicle types to handle their own unique rental rates.
- **Composition:** The Booking class will contain objects of User and Vehicle, creating a relationship-based data structure.
- **Classes & Objects:** We will have a Car class and a Customer class to keep data organized.
- **File Handling:** We will save all car and user data in .txt files so it doesn't disappear when the program closes.

6. Tools Used

Programming Language	C++
Integrated Development Editor	Visual Studio Code

7. Future Scope

- **Database Integration:** Moving from simple text files to a professional database like MySQL or SQLite for handling thousands of vehicles.
- **Graphical User Interface (GUI):** Developing a Windows or Mobile application using Qt or SFML instead of a console-based black screen.
- **GPS Tracking Integration:** Simulating real-time location tracking for each "Rented" vehicle object.
- **Automatic Maintenance Alerts:** Adding a feature that flags a car for "Service" after it has been rented for a specific number of hours.
- **Online Payment Gateway:** Integrating a digital wallet or credit card system to handle the automated billing process.

8. Conclusion

The Car Rental & Dynamic Billing System will serve as a practical demonstration of advanced programming principles. It will not only test our understanding of Object-Oriented Programming (OOP), such as classes and inheritance, but also encourage teamwork, planning, and debugging skills essential for professional software engineering.

By automating the rental process and calculating bills based on real-time data, this project provides a realistic solution to fleet management. This project aligns with the core goals of the Object-Oriented Programming course, bridging theoretical concepts like encapsulation and polymorphism with real-world problem-solving through efficient C++ code.